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# Arboricultural Impact Assessment

# Proposed Haul Road

# Swindon

June 2022

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## 1.0 INTRODUCTION

1.1 This report has been prepared by FPCR Environment and Design Limited on behalf of Hannick, Hallam and Taylor Wimpey to provide an Arboricultural Impact Assessment and details of those trees and hedgerows which would require removal and/or pruning to facilitate the proposed haul road forged off Thornhill Road, South Marston, East of Swindon (hereafter referred to as 'the site').

1.2 The site is centered on Ordnance Survey grid reference SU 195 872.

1.3 The survey was carried out on 20th January 2022.

### Scope of Assessment

1.4 The tree survey and assessment of existing trees has been carried out in accordance with guidance contained within British Standard 5837:2012 *'Trees in Relation to Design, Demolition and Construction – Recommendations'* (hereafter referred to as BS5837). The guidelines set out a structured assessment methodology to assist in determining which trees would be deemed either as being suitable or unsuitable for retention.

1.5 The guidance also provides recommendations for considering the relationship between existing trees and how those trees may integrate into designs for development; demolition operations and future construction processes so that a harmonious and sustainable relationship between any retained trees and built structures can be achieved.

1.6 The purpose of the report is therefore to firstly, present the results of an assessment of the existing trees' arboricultural value, based on their current condition and quality and to secondly, provide an assessment of impact arising from the proposed development of the site.

### Planning History

1.7 The site has received planning approval (reference Application Number S/OUT/13/1555) for a large residential development on Land North of A420, South Marston and Rowborough.

1.8 The proposed development includes up to 2380 residential dwellings, mixed use local center/areas (including A1 retail up to 1,500m$^2$, services (A2), restaurants, pubs and takeaways (A3, A4, A5) business uses (B1) up to 1,000m$^2$, residential), community uses (D1), sheltered and/or care accommodation (C2, C3), two primary schools, green infrastructure including formal (including playing fields) and informal open space, landscape, biodiversity and amenity space (including LEPS, NEAPS/MUGA), changing and sports facilities (including D2 and ancillary retail uses), sustainable drainage systems; highway, cycle and pedestrian routes; car parking; infrastructure (including utilities); engineering works including ground remodeling; demolition, site reclamation and removal of structures; the formation of new accesses from the A420, Old Vicarage Lane and Thornhill Road , highway, cycle and pedestrian routes.

### Purpose of the Summary

1.9 As part of the Conditional Outline Approval, detailed designs for the primary streets will be required. Prior to this a haul road (that follows the same route) will allow construction access to phase 1 South Marston. This Arboricultural Impact Assessment relates to the proposed Haul

Road. The Thornhill Road and Old Vicarage Lane access work areas are covered by separate AIAs.

1.10 The details provided within this Arboricultural Impact Assessment are also required to satisfy two tree related planning conditions of the Outline Planning Application, namely C12 and C13.

## 2.0 PLANNING POLICY

**National Planning Policy Framework July 2021**

2.1 National Planning Policy is defined by the National Planning Policy Framework (NPPF). This sets out the Government's most current and up to date planning policies for England and how these should be applied. The current NPPF is dated July 2021.

2.2 Paragraphs 10 and 11 of the NPPF state that there is a presumption in favour of sustainable development and states that for decision making, the LPA should be '*c) approving development proposals that accord with an up-to-date development plan without delay'.* In the absence of a development plan or the development plan is out of date, the acting LPA should grant planning consent so far as the development proposals do not breach the policies and guidance outlined in the NPPF.

2.3 In relation to arboriculture, the NPPF also states that:

- *131 'Trees make an important contribution to the character and quality of urban environments, and can also help mitigate and adapt to climate change. Planning policies and decisions should ensure that new streets are tree-lined (footnote 50), that opportunities are taken to incorporate trees elsewhere in developments (such as parks and community orchards), that appropriate measures are in place to secure the long-term maintenance of newly-planted trees, and that existing trees are retained wherever possible. Applicants and local planning authorities should work with highways officers and tree officers to ensure that the right trees are planted in the right places, and solutions are found that are compatible with highways standards and the needs of different users' (footnote 50: unless, in specific cases, there are clear, justifiable and compelling reasons why this would be inappropriate)*
- *180 I 'development resulting in the loss or deterioration of irreplaceable habitats (such as ancient woodland and ancient or veteran trees) should be refused, unless there are wholly exceptional reasons (footnote 63) and a suitable compensation strategy exists';*

  and provides specific guidance that:

- *180 (d) 'development whose primary objective is to conserve or enhance biodiversity should be supported; while opportunities to incorporate biodiversity improvements in and around developments should be encouraged, especially where this can secure measurable net gains for biodiversity'.*

2.4 With reference to paragraph 180 (c), examples of what is deemed to be *'wholly exceptional'* are included within Footnote 63 and provides the examples of '*infrastructure projects (including nationally significant infrastructure projects, orders under the Transport and Works Act and hybrid bills), where the public benefit would clearly outweigh the loss or deterioration of habitat'.*

### Statutory Considerations

2.5 Local authorities have a Duty under the Town and Country Planning Act to create Tree Preservation Orders (TPO) in order to protect and preserve specific trees and woodlands that bring significant amenity benefit to a particular site or location. Under a TPO it is a criminal offence to cut down, top, lop, uproot or wilfully destroy a tree protected by that Order, or to cause or permit such actions, if carried out without the prior written consent of the acting LPA. Anyone found guilty of such an offence is liable and in serious cases, may result in prosecution and incur an unlimited fine.

2.6 Following consultation with the Local Planning Authority, Swindon Borough Council, it is understood that there is a Tree Preservation Order, namely TPO 2 / 2015, which applies to a number of trees present within the Proposed Haul Road route. For specific tree details, see table 2.

2.7 Information provided on Tree Preservation Orders and Conservation Areas is accurate to the date of this assessment and cannot be assumed to remain unchanged. The last check was carried out on the 26.02.2015.

### Non-Statutory Considerations

2.8 In order to compile existing baseline information on relevant arboricultural considerations information was requested from both statutory and non-statutory nature conservation organisations. The Multi Agency Geographic Information for the Countryside (MAGIC)[1] website highlighted no tree cover within the Land East of Thornhill Road site as or included within the following:

- The Priority Habitat Inventory, Deciduous Woodland
- The National Forestry Inventory

## 3.0 SURVEY METHODOLOGY

3.1 The survey of trees has been carried out in accordance with the criteria set out in Chapter 4 of BS5837. The survey has been undertaken by a suitably qualified and experienced arboriculturist and has recorded information relating to all those trees within the site and those adjacent to the site which may be of influence to any proposals. Trees were assessed for their arboricultural quality and benefits within the context of the proposed development in a transparent, understandable and systematic way.

3.2 Due to the time that has lapsed since the original tree survey in 2017, a resurvey of all the trees and hedgerows within the vicinity the proposed haul route are identified, reviewing their condition and updating previous survey information accordingly. The resurvey of trees was undertaken in January 2022. This has ensured that all the trees and hedgerows within the vicinity of the proposed route which would be affected by the works have been identified.

3.3 Trees have been assessed as groups or hedgerows where it has been determined appropriate.

[1] http://magic.defra.gov.uk/

- The term group has been applied where trees form cohesive arboricultural features either aerodynamically, visually or culturally including biodiversity or habitat potential for example parkland or wood pasture.
- For the purposes of this assessment, a hedgerow is described as any boundary line of trees or shrubs less than 5m wide at the base and are managed under a regular pruning regime.

3.4 An assessment of individual trees within groups or hedgerows has been made where a clear need to differentiate between them, for example, in order to highlight significant variation between attributes including physiological or structural condition or where a potential conflict may arise.

### Ancient and Veteran Trees

3.5 Veteran trees and Ancient Woodland are important components of the landscape, their importance can be for a number of reasons including that of their ecological, social, cultural and historic value.

3.6 Veteran Trees and Ancient Woodlands are material considerations within the planning process and their importance is specifically recognised within the National Planning Policy Framework (NPPF) 2021, which defines the terms ancient or veteran tree as:

*'A tree which, because of its age, size and condition, is of exceptional biodiversity, cultural or heritage value. All ancient trees are veteran trees. Not all veteran trees are old enough to be ancient, but are old relative to other trees of the same species. Very few trees of any species reach the ancient life-stage.'*[2]

3.7 Various published methodologies are currently available which, due to the complexity and subjectivity of the process of defining and assessing these trees, often have conflicting definitions. This assessment, and the criteria used for defining ancient/veteran trees and the identification of attributable ancient/veteran features, has been based on a range of currently published guidance and resources.

### BS5837 Categories

3.8 Trees have been divided into one of four categories based on Table 1 of BS5837, *'Cascade chart for tree quality assessment'*. For a tree to qualify under any given category it should fall within the scope of that category's definition (see below).

3.9 Category U trees are those which would be lost in the short term for reasons connected with their physiology or structural condition. They are, for this reason not considered in the planning process on arboricultural grounds. Categories A, B and C are applied to trees that should be of material considerations in the development process. Each category also having one of three further sub-categories (i, ii, iii) which are intended to reflect arboricultural, landscape and cultural or conservation values accordingly.

3.10 **Category (U) – (Red):** Trees which are unsuitable for retention and are in such a condition that they cannot realistically be retained as living trees in the context of the current land use for longer than 10 years. Trees within this category are:

[2] Ministry of Housing, Communities and Local Government. (2019). *National Planning Policy Framework.* London: Ministry of Housing, Communities and Local Government.

- Trees that have a serious irremediable structural defect such that their early loss is expected due to collapse and includes trees that will become unviable after removal of other category U trees.
- Trees that are dead or are showing signs of significant, immediate or irreversible overall decline.
- Trees that are infected with pathogens of significance to the health and/ or safety of other nearby trees or are very low quality trees suppressing adjacent trees of better quality.
- Certain category U trees can have existing or potential conservation value which may make it desirable to preserve.

3.11 **Category (A) – (Green):** Trees that are considered for retention and are of high quality with an estimated remaining life expectancy of at least 40 years with potential to make a lasting contribution. Such trees may comprise:

- Sub category (i) trees that are particularly good examples of their species, especially if rare or unusual, or are essential components of groups such as formal or semi-formal arboricultural features for example the dominant and/or principal trees within an avenue.
- Sub category (ii) trees, groups or woodlands of particular visual importance as arboricultural and / or landscape features.
- Sub category (iii) trees, groups or woodlands of significant conservation, historical, commemorative or other value for example veteran or wood pasture.

3.12 **Category (B) – (Blue):** Trees that are considered for retention and are of moderate quality with an estimated remaining life expectancy of at least 20 years with potential to make a significant contribution. Such trees may comprise:

- Sub category (i) trees that might be included in category A but are downgraded because of impaired condition for example the presence of significant though remediable defects, including unsympathetic past management and storm damage.
- Sub category (ii) trees present in numbers, usually growing as groups or woodlands, such that they attract a higher collective rating than they might as individuals or trees occurring as collectives but situated so as to make little visual contribution to the wider locality.
- Sub category (iii) trees with material conservation or other cultural value.

3.13 **Category (C) – (Grey):** Trees that are considered for retention and are of low quality with an estimated remaining life expectancy of at least 10 years or young trees with a stem diameter below 150mm. Such trees may comprise:

- Sub category (i) unremarkable trees of very limited merit or such impaired condition that they do not qualify in higher categories.
- Sub category (ii) trees present in groups or woodlands, but without this conferring on them significantly greater collective landscape value or trees offering low or only temporary / transient screening benefits.
- Sub category (iii) trees with no material conservation or other cultural value.

### Site Plans

3.14 The individual positions of trees and groups have been shown on the Tree Survey Plan. The positions of trees are based on a topographical / land survey, as far as possible, supplied by the client. Where topographical information has not identified the position of trees these have been plotted using a global positioning system and aerial photography to provide approximate locations. The crown spread, root protection area and shade pattern (where appropriate) are also indicated on this plan.

3.15 As part of this assessment, a Tree Retention Plan (1796-T-13, 1796-T-14, 1796-T-15) has been prepared to show the proposed layout in relation to the existing tree cover allowing an assessment of any potential conflicts. The plan also identifies which trees would be required to be removed or retained as part of the proposed development.

3.16 A Tree Protection Plan (1796-T-16, 1796-T-17, 1796-T-18) has also been included which outlines proposed tree protection measures to be enacted during the construction phase.

### Tree Constraints and Root Protection Areas

3.17 Below ground constraints to future development are represented by tree roots and the soil environment in which they grow which needs to be protected if the tree is to be retained. Tree rooting systems are essential for the uptake of water and nutrients, serving the storage of carbohydrates for the future growth and function of the tree, and form structural anchorage and support for the stem and crown. The perceived rooting area of the tree; referred to as the root protection area (RPA) needs to be protected if the tree is to be retained.

3.18 The RPA is a notional area considered to be the minimum zone that must be protected to avoid any adverse impacts on retained trees. The RPA has been calculated in accordance with Annex C, D and Section 4.6 of BS5837:2012 and requires suitable protection in order for the tree to be successfully incorporated into any future scheme. As such, the RPA of existing trees is an important material consideration when considering site constraints and planning development activities.

3.19 Where applicable the shape of the Root Protection Area has been modified to consider the presence of any nearby obstacles (existing or past) which may have restricted root growth and the likely root distribution i.e. the presence of hard standing, structures and underground apparatus. Where groups of trees have been assessed, the Root Protection Area has been shown based on the maximum sized tree in any one group and so may exceed the Root Protection Area required for some of the individual specimens within the group. Further detailed inspection of the individual trees forming a group may be required where development impacts upon the group.

3.20 Whilst it is generally accepted that a trees roots may extend far greater distances than the notional RPA, with the distribution of the root system relating directly to the availability of suitable conditions for growth (namely oxygen, water and nutrients), with roots predominantly located in the upper 1,000 mm of the soil horizon; the RPA offers an accepted protective buffer from development.

3.21 Above ground constraints such as the current crown spread of the trees and an illustration of the shade pattern (where appropriate) have been considered and identified within the Tree Survey Plan and Tree Retention Plan indicates their potential area of shading influence.

### Considerations and Limitations of the Tree Survey

3.22 The survey was completed from ground level only and from within the boundary of the site. Aerial tree inspections or an assessment of the internal condition of the stem/s or branches were not undertaken at this stage as this level of survey is beyond the scope of the initial assessment.

3.23 The statements made in this report regarding the assessed trees does not take into account the effects of extreme / adverse weather conditions, changes in land use prior to the site's development, unforeseen accidents or anti-social behaviours, such as vandalism, which occur since the date of the survey. As such, the assessment of tree condition given within applies to the date of survey and cannot be assumed to remain unchanged.

3.24 It will be necessary to review all comments and observations made within this report, in accordance with sound arboricultural practice, within two years of the date of survey (unless explicitly stated elsewhere within this report). Further review may also be necessary where site conditions change or works to trees are carried out which have not been specified in detail within this report.

3.25 Hedgerows are identified as a Habitat of Principal Importance (HPI) as listed within Section 41 of the Natural Environment and Rural Communities (NERC) Act 2006. The tree survey conducted, in accordance with BS5837, does not assess hedgerows against the Hedgerow Regulations 1997 or specifically from an ecological perspective, and is outside the scope of this assessment.

3.26 It may be necessary during detailed design to undertake further assessment and accurate positioning of woody species within tree groups and hedgerows to assist structural calculations for foundation design of structures in accordance with current building regulations. The exact position of individual trees or species included as part of a tree group should be checked and verified on site prior to any decisions for foundation design, tree operations or construction activity being undertaken. Further survey work would be required for calculating foundation depths in accordance with NHBC Chapter 4.2 Building near Trees.

## 4.0 RESULTS

4.1 A total of thirty-one individual trees, sixteen groups of trees and twenty one hedgerows were surveyed as part of the Arboricultural Assessment. Trees were surveyed as individual trees and groups of trees where examples are clearly present as per the description. Refer to the Tree Survey Plan and Appendix A – Tree Schedule for full details of the trees included in this assessment. The table below summarises the trees assessed.

### Tree Schedule

4.2 Appendix A provides a schedule of trees and hedgerows for the wider site but only those trees within the vicinity of the Haul Road have been reviewed and information pertaining to those trees updated as required. All other trees subject to the original survey (except for those affected by

the two remaining points of access, covered by separate Arboricultural Impact Assessment Reports) have not been reviewed at this stage and information pertaining to those trees in the Tree Schedule within the wider site remain as per the 2017 Tree Survey.

4.3 Appendix A presents details of any individual trees, groups and hedgerows found during the assessment including heights, diameters at breast height, crown spread (given as a radial measurement from the stem), age class, comments as to the overall condition at the time of inspection, BS5837 category of quality and suitability for retention and the root protection area.

4.4 General observations particularly of structural and physiological condition for example the presence of any decay and physical defect and preliminary management recommendations have also been recorded where appropriate.

### Results Summary

**Table 1: Summary of Trees by Retention Category**

| | Individual Trees | Total | Groups of Trees & Hedgerows | Total |
|---|---|---|---|---|
| Cate–ory U - Unsuitable | T3, T33, T44, T46, T47, T54 | 6 | G3 | 1 |
| Category A (High Quality / Value) | T37 | 1 | | 0 |
| Category B (Moderate Quality / Value | T34, T53, T55, T102, T103 | 3 | G1, G4, G6, G13, G41, G42 | 6 |
| Category C (Low Quality / Value) | T1, T4, T5, T6, T13, T14, T15, T16, T17, T18, T19, T20, T21, T22, T36, T39, T40, T41, T42, T45, T48 | 21 | G5, G14, G15, G16, G17, G26, G27, G28, G29, H1, H2, H3, H4, H5, H6, H7, H9, H12, H17, H18, H19, H27, H33, H34, H36, H37, H38, H39, H48, H51 | 29 |

### Ancient and Veteran Trees

4.5 None of the assessed trees were considered as ancient or veteran trees in accordance with accepted methodologies and guidance.

### Statutory Constraints

4.6 The following table details which trees are included in the Swindon Borough Council Tree Preservation Order (TPO), TPO 2 / 2015. The trees identified within the TPO are protected by law from felling or uprooting, pruning including ‘topping/lopping’ and willful damage or destruction. The granting of full planning permission would override the protection afforded by the Tree Preservation Order to those trees shown as removed to facilitate the proposals within the approved plans.

**Table 2: Tree Preservation Order Details**

| Tree No. taken from FPCR | TPO reference no. |
|---|---|
| T34 | T9 |
| T42 | T18 |
| T53 | T19 |
| T54 | T29 |
| T55 | T28 |

4.7 Prior to any tree surgery and / or felling of protected trees it will be necessary to apply to the relevant local planning authority to gain consent for the works. For more information regarding Conservation Areas and Tree Preservation Orders it is advised that contact is made with the Local Planning Authority's arboricultural officer, or other such relevant person.

## 5.0 ARBORICULTURAL IMPACT ASSESSMENT

5.1 The following paragraphs present a summary of the tree survey and discussion of particular trees and groups recorded in the context of any proposed development in the form of an Arboricultural Impact Assessment in accordance with section 5.4 of BS5837. Any final tree retentions will need to be reconciled with the advice contained within this report.

5.2 The AIA has been based upon the Drawing Title, '1335-HR-101 D' produced by Brookbanks and seeks to outline the relationship between the proposals and the existing trees, groups and hedgerows. The drawing shows the proposals for haul road through the site with the access forged off Thornhill Road. An overlay of the layout has been incorporated in the Tree Retention (FPCR 1796-T-13 – 15) and Tree Protection Plans (FPCR1796-T-16 – 18) to assist in identifying the relationship and any potential conflicts between the proposals and the existing trees and hedgerows

**Table 3: Summary of Impact on Tree Stock**

| | Trees to be Retained as part of the Haul Road Application | Total | Trees to be Removed in full or part for the Haul Road Application | Total |
|---|---|---|---|---|
| Category U (Unsuitable) | T3, T33, T44, T46, T47 | 5 | T33, T46, T47, T54, G3 | 5 |
| Category A (High Quality / Value) | T37 | 1 | | 0 |
| Category B (Moderate Quality / Value | T34, T102, T103, G1, G4, G6, G13, G41, G42 | 9 | T34, T53, T55, T56 | 4 |
| Category C (Low Quality / Value) | T1, T4, T5, T6, T13, T14, T15, T16, T17, T18, T20, T21, T22, T36, T39, T40, T41, T42, T45, T48, G5, G14, G15, G16, G17, G27, G28, G29, H5, H6, H7, H9, H12, H18, H19, H27, H34, H36, H37, H38, H39, H51 | 42 | T19, G28, H1, H2, H3, H4, H17, H18, H27, H33, H34, H36, H38 | 13 |

5.3 The Haul Road proposals will necessitate the removal of the following:

**Trees**

5.4 Nine trees require removal to facilitate the proposals, four category U, four category B and one category C.

5.5 T33 was a common ash, which showed signs of extensive decline during the resurvey and as such would be considered as category U by virtue of its poor condition.

5.6 T46 was a crack willow, the main leader of which has failed into the field to the northwest of the stem location. For reduced structural integrity and a safe useful life expectancy which is unlikely to extend beyond 10 years, it was considered a category U specimen and unsuitable for retention.

5.7 T47 was another crack willow, the whole stem of which had failed. For this reason, it was also regarded to be a category U specimen and unsuitable for retention.

5.8 T54 was a category U English oak due to its declining form, significant basal decay and buttress root damage on the southwestern side. Dieback within the crown was considered major due to the majority of the upper crown being dead. Major bark delamination was also present on the south and west aspects for almost the full height of the stem.

5.9 T34 was a category B English oak *Quercus robur,* subject of a Tree Preservation Order. There was visible dead wood within the crown resulting in early onset peripheral crown dieback which is highly likely to be attributable to waterlogged ground conditions over time.

5.10 T53, T55 and T56 were category B English oaks located within H33 along the southern side of Old Vicarage Lane. These specimens also have Tree Preservation Orders upon them. T54 has minor deadwood present, T55 showed visible decline within the canopy, especially the upper

eastern aspect. Storm damage and major deadwood was present. T56 also showed signs of historic storm damage and had major deadwood focused on the northern aspect.

5.11 Finally, T19, a category C crack willow with a large quantity of deadwood within the crown.

**Groups**

5.12 Two groups require partial removal to facilitate the proposals, one category U & one Category C.

5.13 G3, a category U group consisted of a single apple *Malus domestica* specimen and two dead standing stemmed pears *Pyrus calleryana.* The apple was in extensive decline with severe basal damage and decay present with two *Inonotus hispidus* fungal brackets present within upper branches.

5.14 G28 was comprised of eight crack willow *Salix fragilis,* most of which were old pollards. Their overall condition was ranging from fair to poor, both structurally and physiologically. Several of the stems had collapsed and there was evidence of wind loaded damage to branches. Two trees from within this group would need to be removed to facilitate the proposals.

**Hedges**

5.15 Eleven hedgerows will require partial removal to facilitate the haul road proposals, all of which were category C. The average width of these removals would be in the region of 18 – 20 metres.

5.16 H1 H4 were category C field boundary hedgerows which lacked any targeted management. As such, they were outgrown with a number of dead specimens throughout.

5.17 From an arboricultural perspective, H2 was a low quality internal field boundary hedgerow. It had been historically managed to maintain form and shape.

5.18 H3 was also a low arboricultural quality internal field boundary hedgerow and was growing along the line of a ditch course. There were some self-seeded sycamore *Acer pseudoplatanus* and common ash *Fraxinus excelsior* visible in places, which had attained up to 7m in height. The overall height of the hedgerow was c. 5m.

5.19 H17, an internal field boundary hedgerow, was growing along the edge of a ditch course. Throughout the hedge there was an abundance of ivy and the hedgerow presented a ‘gappy’ form, with a number of dead elm.

5.20 H18, a further low arboricultural quality internal field boundary hedgerow supported a high amount of ivy throughout. It was of well-established form although showed limited signs of previous management.

5.21 H27, an internal field boundary hedgerow supported numerous dead elm throughout and due to absence of management had become defunct in places.

5.22 H33 was also a low-quality hedgerow from an arboricultural perspective running along the roadside. At the time of survey, it had not been maintained for several years and had vertical growth to 6m. There were several dead elm specimens present throughout the hedgerow.

5.23 H34 was an internal field boundary hedgerow. It had an unmanaged, outgrown form and contained several gaps along its length.

5.24 H36, was an internal field boundary hedgerow. There were several dead elm present and there was little signs of recent management. In the past, there were signs that the hedgerow had been regularly managed in the past to maintain a consolidated shape and form.

5.25 H38, another internal field boundary hedgerow which had been regularly managed to maintain a consolidated form.

### Discussion

5.26 In conclusion for arboriculture, the proposals are considered to meet the aims and objectives of national policy through careful consideration of the design and the seeing retention of as much of the existing tree and hedgerow cover as possible whilst allowing for the haul road access. The wider development proposals will provide extensive new landscaping including tree and hedgerow planting around all of the new access point thus mitigating the loss of tree and hedgerow cover to the proposals.

### Tree Management

5.27 All retained trees should be subjected to sound arboricultural management as recommended within section 8.8.3 of BS5837 *Post Development Management of Existing Trees,* where there is a potential for public access in order to satisfy the landowner's duty of care. Additionally, inspections annually and following major storms should be carried out by an experienced arboriculturist or arborist to identify any potential public safety risks and to agree remedial works as required.

5.28 All tree works undertaken should comply with British Standard 3998:2010 and should therefore be carried out by skilled tree surgeons. It would be recommended that quotations for such work be obtained from Arboricultural Association Approved Contractors as this is the recognised authority for certification of tree work contractors.

5.29 All vegetation and, particularly, woody vegetation proposed for clearance should be removed outside of the bird-breeding season –March - September inclusive) as all birds are protected under the Wildlife and Countryside Act, 1981 (as amended) whilst on the nest. Where this is not possible, vegetation should be checked for the presence of nesting birds prior to removal by an experienced ecologist.

### General Design Principles in Relation to Retained Trees

5.30 The routing of below ground services should also be considered with regard to the retained trees. As recommended by the guidance given in section 7.7 of BS5837 services, where possible, should not encroach within the Root Protection Areas of retained trees. If below-ground services are proposed within a Root Protection Area, modifications to the alignment of the service route may need to be made in order to minimise adverse effects on root stability and overall tree health.

5.31 Consideration may also need to be given to the potential for tree roots of newly planted trees and hedgerows to affect or compromise the future services. As far as feasible, it would be preferable that proposed services near both the existing and any new planting should be ducted for ease of access and maintenance and grouped together to minimise any future disturbance.

## 6.0 NEW TREE AND HEDGEROW PLANTING

6.1 As part of the development proposals new tree planting will be proposed which will more than mitigate the losses required for the Haul Road proposals.

6.2 As part of the mitigation for the losses of trees, it would be proposed that replacement planting would be included within public open space within the wider development.

6.3 Species composition and locations for all trees, including those replacement trees for TPO losses, will be determined within future landscaping proposals.

6.4 Where Tree Preservation Order trees are removed, the replacement of these trees will be agreed and approved by the LPA as part of the wider landscaping proposals.

### Trees

6.5 The landscaping scheme should consider the use of both native tree species (for their low maintenance requirements and nature conservation value) and ornamental species (for their contribution to urban design and amenity value). Species choices should be selected on the basis of their suitability for the final site use. Furthermore, during the design process consultation should be made with the Local Planning Authority to obtain information on their tree strategy and incorporate the planting proposals with any local policies and initiatives and/or Biodiversity Action Plans (BAP).

6.6 In line with the NPPF all schemes should aim achieve a net gain in biodiversity value. Nationally recognised biodiversity metrics allow for the inclusion of, not limited to, newly planted scattered trees, woodlands and hedgerows as a means of compensating for loss of habitat as part of the development. Tree and shrub planting can therefore be used to contribute to this biodiversity gain.

6.7 To maximise biodiversity value (and contribution to net gain) native species or varieties should be specified. Such provisions can be incorporated into both the hard and soft landscaping of the scheme. It is recommended that tree and hedgerow specifications are made following consultation with guidance published by the Local Planning Authority.

6.8 When deciding upon suitable tree species, careful consideration would need to be given to the following: ultimate height and canopy spread, form, habit, density of crown, potential shading effect, colour, water demand, soil type and maintenance requirements in relation to both the built form of the new development and existing properties.

6.9 Through careful species selection, the landscape scheme shall reduce the risk of trees being removed in the future on the grounds of nuisance. Nuisance can be perceived in a number of ways and vary from person to person however most commonly, within the context of trees, low overhanging branches, excessive shading, seasonal leaf fall and the misinformed perception that trees close to buildings cause damage.

### Hedgerows

6.10 Hedgerows are identified as a Habitat of Principle Importance (HPI) as listed within Section 41 of the Natural Environment and Rural Communities (NERC) Act 2006. Consequently, it is important

that the proposed scheme delivers a net gain in terms of linear hedgerows through new planting to compensate for any losses. Species should be native, and characteristic of the locality.

6.11 Recommended species for native hedgerow planting are as follows:

- *Crataegus monogyna*
- *Prunus spinosa*
- *Cornus sanguinea*
- *Corylus avellana*
- *Acer campestre*
- *Euonymus europaeus*

**General Planting Recommendations**

6.12 Wherever possible, following discussions with the developer and utility companies, common service trenches should be specified to minimise land take associated with underground service provision and facilitation access for future maintenance.

6.13 Tree planting should be avoided where they may obstruct overhead power lines or cables. Any underground apparatus should be ducted or otherwise protected at the time of construction to enable trees to be planted without resulting in future conflicts.

## 7.0 TREE PROTECTION MEASURES

7.1 Retained trees will be protected from construction work by Tree Protection Fencing as shown in the Tree Protection Plan (FPCR 1796-T-16 - 18) and in accordance with BS5837 (2012). Appendix B provides the specification of tree protective fencing that will be used.

**General Information and Recommendations**

7.2 All trees retained on site will be protected by suitable barriers or ground protection measures around the calculated RPA, crown spread of the tree or other defined constraints of this assessment as detailed by section 6 and 7 of BS5837.

7.3 Barriers will be erected prior to commencement of any construction work and before demolition including erection of any temporary structures. Once installed, the area protected by fencing or other barriers will be regarded as a construction exclusion zone. Fencing and barriers will not be removed or altered without prior consultation with the Project Arboriculturist.

7.4 Any trees that are not to be retained as part of the proposals should be felled prior to the erection of protective barriers. Particular attention needs to be given by site contractors to minimise damage or disturbance to retained specimens.

7.5 Where it has been agreed, construction access may take place within the root protection area if suitable ground protection measures are in place. This may comprise single scaffold boards over a compressible layer laid onto a geo-textile membrane for pedestrian movements. Vehicular

movements over the root protection area will require the calculation of expected loading and the use of proprietary protection systems.

7.6 Confirmation that tree protective fencing or other barriers have been set out correctly should be gained prior to the commencement of site activity.

**Tree Protection Barriers**

7.7 Tree protection fencing should be fit for the purpose of excluding any type of construction activity and suitable for the degree and proximity of works to retained trees. Barriers must be maintained to ensure that they remain rigid and complete for the duration of construction activities on site.

7.8 In most situations, fencing should comprise typical construction fencing panels attached to scaffold poles driven vertically into the ground. For particular areas where construction activity is anticipated to be of a more intense nature, supporting struts, acting as a brace should be added and fixed into position through the application of metal pins driven into the ground to offer additional resistance against impacts.

7.9 Where site circumstances and the risk to retained trees do not necessitate the default level of protection an alternative will be specified appropriate to the level / nature of anticipated construction activity. The recommended methods of fencing specifications for this site have been illustrated in Appendix B.

7.10 It may be appropriate on some sites to use temporary site offices, hoardings and lower level barrier protection as components of the tree protection barriers. Details of the specific protection barriers for the site can be provided should the application be approved, as part of a site specific Arboricultural Method Statement for a Reserved Matters application and in accordance with the guidance contained within BS5837.

**Protection outside the exclusion zone**

7.11 Once the areas around trees have been protected by the barriers, any works on the remaining site area may be commenced providing activities do not impinge on protected areas.

7.12 All weather notices should be attached to the protective fencing to indicate that construction activities are not permitted within the fenced area. The area within the protective barriers will then remain a construction exclusion zone throughout the duration of the construction phase of the proposed development. Protection fencing signs can be provided upon request.

7.13 Wide or tall loads etc should not come into contact with retained trees. Banksman should supervise transit of vehicles where they are in close proximity to retained trees.

7.14 Oil, bitumen, cement or other material that is potentially injurious to trees should not be stacked or discharged within 10m of a tree stem. No concrete should be mixed within 10m of a tree. Allowance should be made for the slope of ground to prevent materials running towards the tree.

7.15 No fires will be lit where flames are anticipated to extend to within 5m of tree foliage, branches or trunk, taking into consideration wind direction and size of fire.

7.16 Notice boards, telephone cables or other services should not be attached to any part of a retained tree.

7.17 Any trees which need to be felled adjacent to or are present within a continuous canopy of retained trees, must be removed with due care (it may be necessary to remove such trees in sections).

**Protection of Trees Close to the Site**

7.18 A number of trees were located on the boundaries of the site and therefore the root protection area and crown spread of these trees will need to be protected in the same way as all the retained trees within the site. All trees located outside the boundaries of the assessment site yet within close proximity to works should be adequately protected during the course of the development by barriers or ground protection around the calculated root protection area.

7.19 Any trees which are to be retained and whose Root Protection Areas may be affected by the development should be monitored, during and after construction, to identify any alterations in quality with time and to assess and undertake any remedial works required as a result.

**Protection for Aerial Parts of Retained Trees**

7.20 Where it is deemed necessary to operate wide or tall plant within close proximity to trees it is best advised that appropriate, but limited tree surgery, be carried out beforehand to remove any obstructive branches as any such equipment would have potential to cause damage to parts of the crown material, i.e. low branches and limbs, of retained trees within the protective barriers. This is termed as 'access facilitation pruning' within BS5837. Any such pruning should be undertaken in accordance with a specification prepared by an arboriculturist.

7.21 A pre-commencement site meeting with contractors who are responsible for operating machinery is advised to firstly highlight the potential for damage occurring to tree crowns and to ensure that extra care is applied when manoeuvring machinery during such operations within close proximity to retained trees to avoid any contact.

7.22 In the event of having caused any branch or limb damage to retained trees it is strongly recommended that suitable tree surgery be carried out, in accordance with British Standard 3998:2010 and in agreement with the Local Planning Authority prior to correcting the damage, upon completion of development.

## 8.0 CONCLUSION

8.1 The Haul Road will require the removal of nine individual trees, two individuals within a tree group, the complete removal of one category U group and sections from eleven hedgerows to facilitate the proposals.

8.2 Whilst the Haul Road will require the loss of trees and hedgerows, they will be more than mitigated for through new high-quality tree and hedgerow planting across the wider development as part of the extensive Green Infrastructure.

8.3 Where the four trees with TPO placed upon them are to be removed, the manner of their replacement should be discussed and agreed with the LPA.

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## FIGURES

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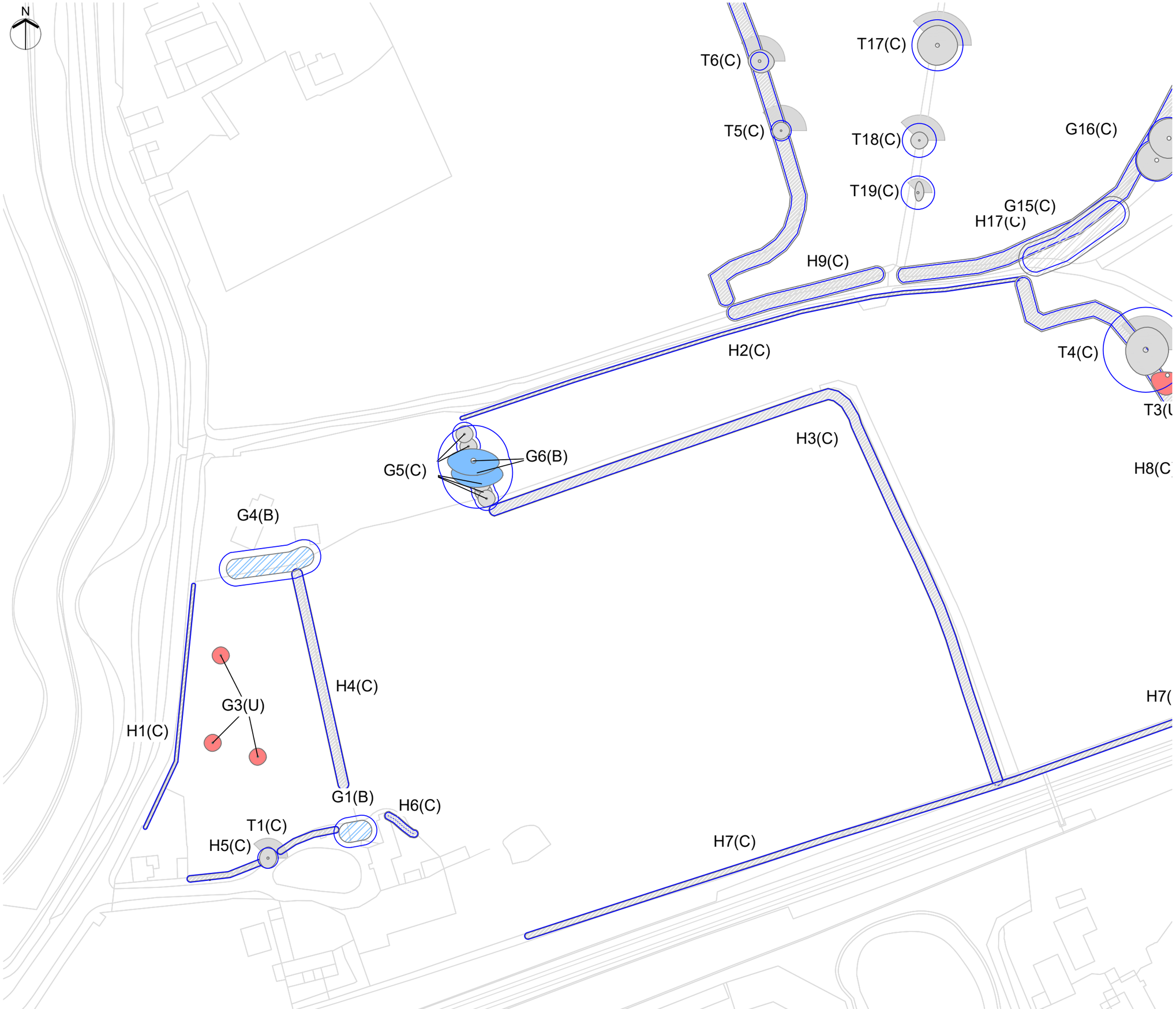


## KEY

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- Category A - Trees / Groups of High Quality (BS 5837:2012)
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Scale 1:1250 @ A3

0 25 50 75m

NOTES

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Taylor Wimpey

project
Land North of A420 South Marston and Rowborough, Swindon

drawing title
TREE SURVEY PLAN
Haul Road

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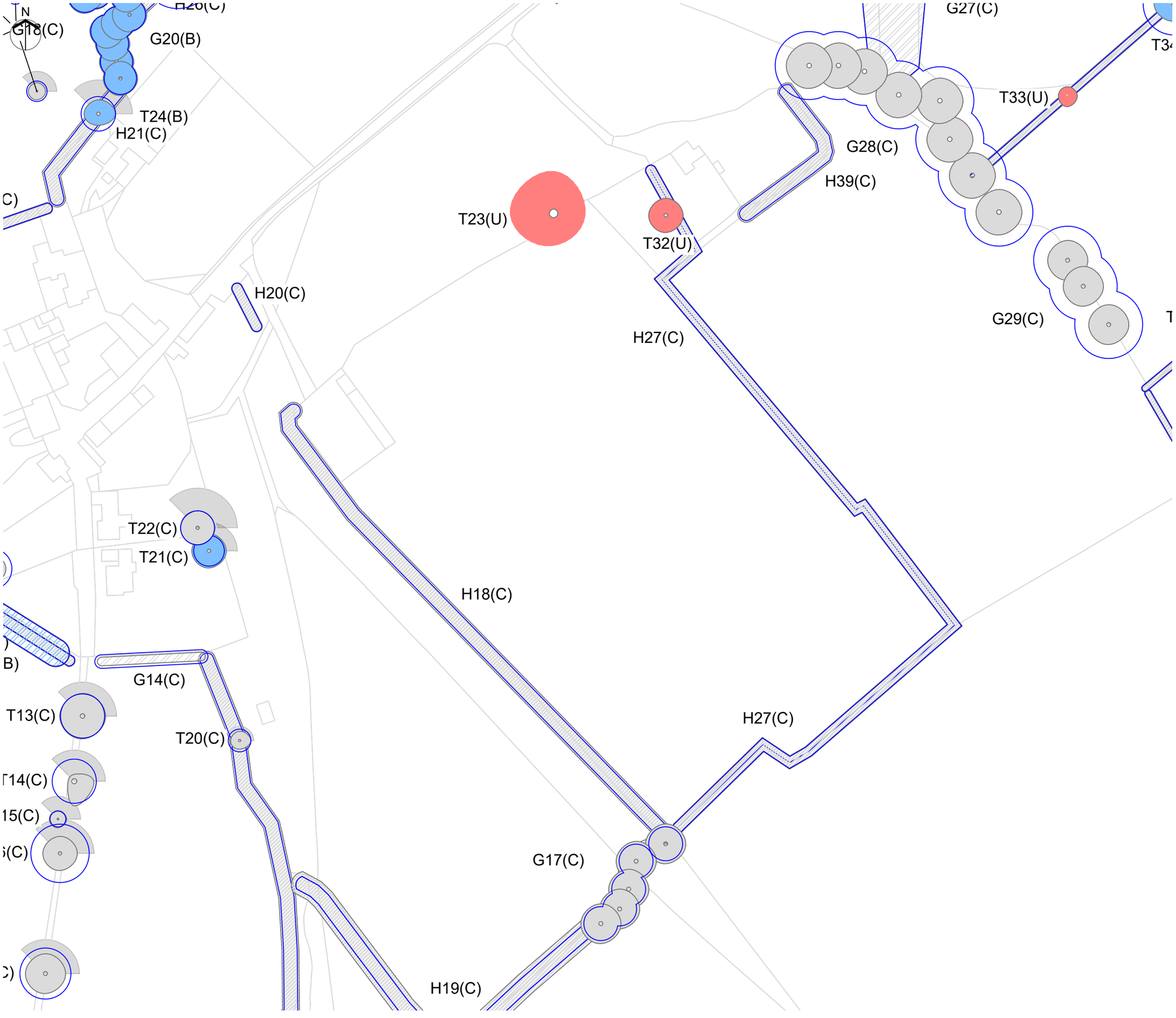


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drawing title
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Haul Road

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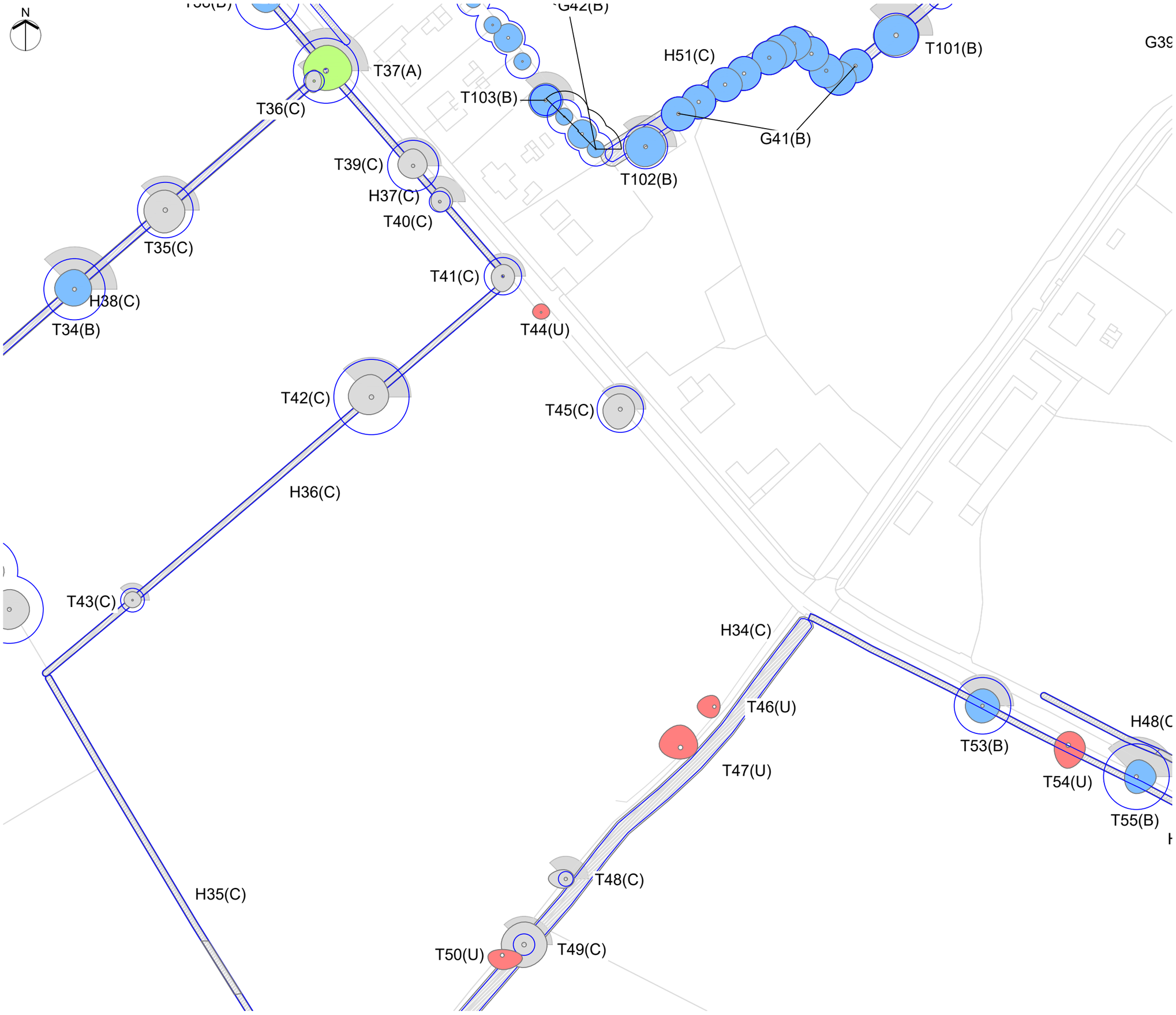


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Haul Road

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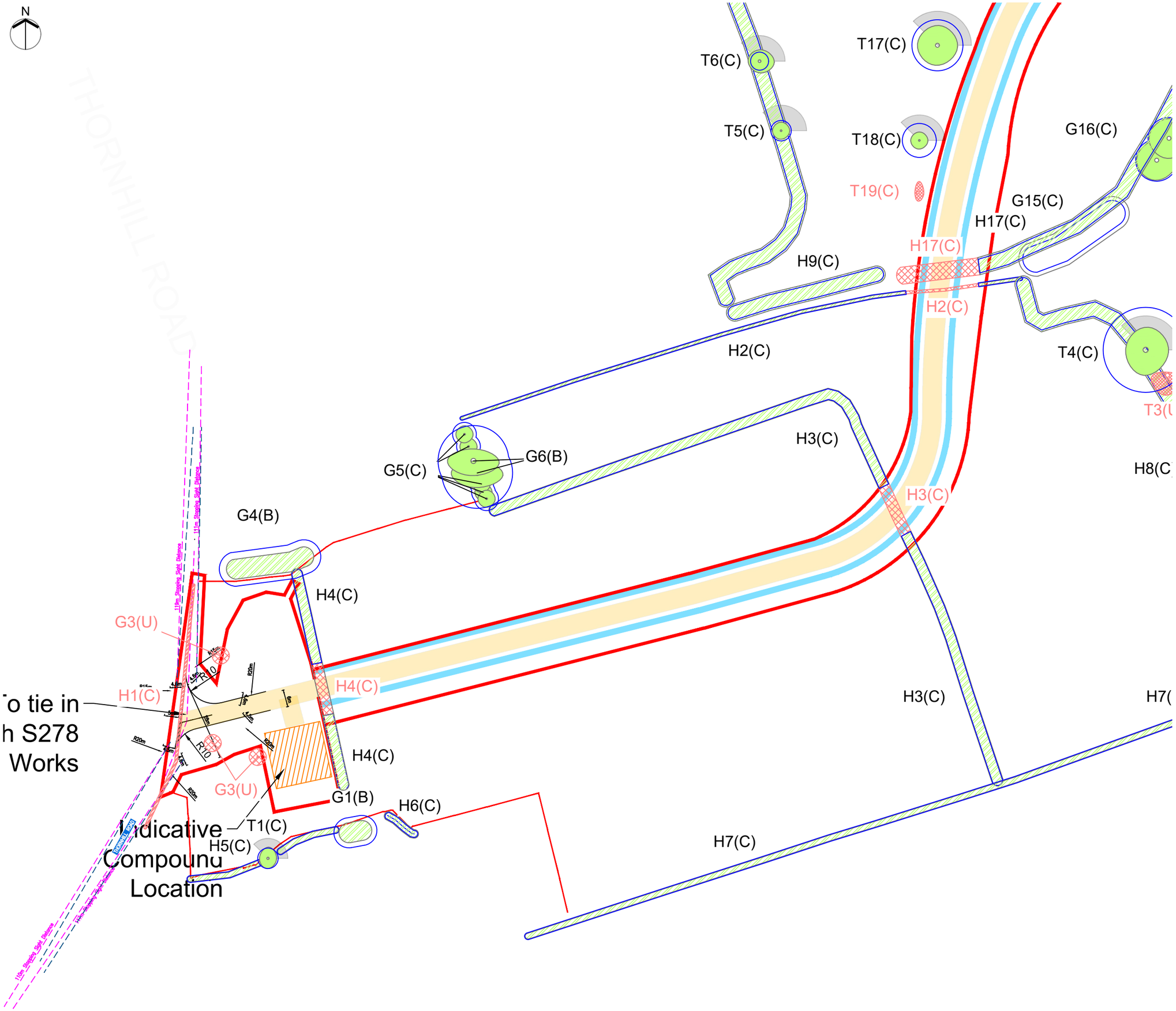


## KEY

- Tree/Group to be Retained
- Tree/Group proposed to be removed subject to relevant permissions
- Category U - Unsuitable for retention on arboricultural grounds
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0 25 50 75m

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**1796-T-13**

rev
**E**

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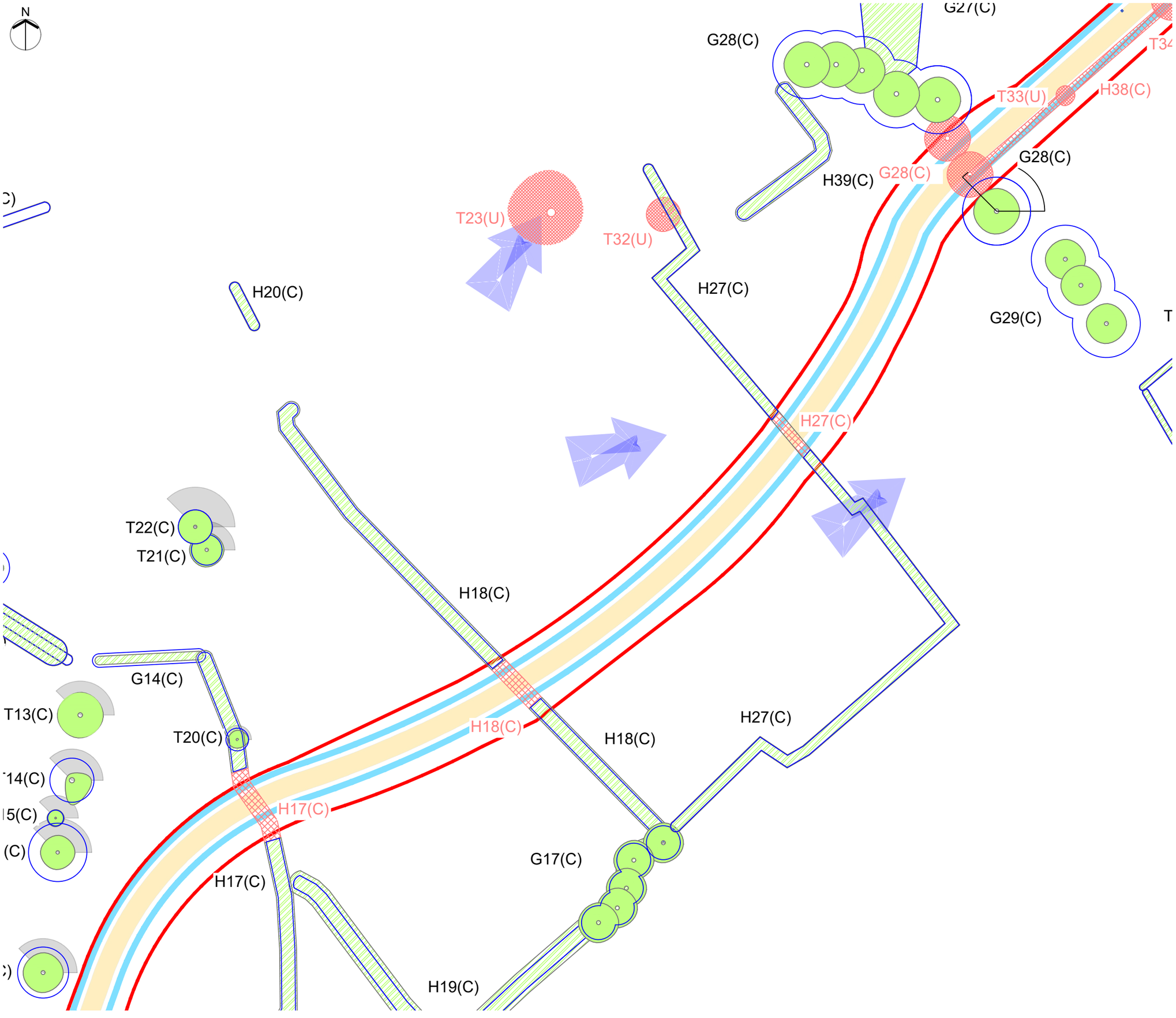


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| D | 27.05.2022 | Layout Update - 1335-HR-101 D | MHE |
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Haul Road

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drawing number **1796-T-14** rev **E**

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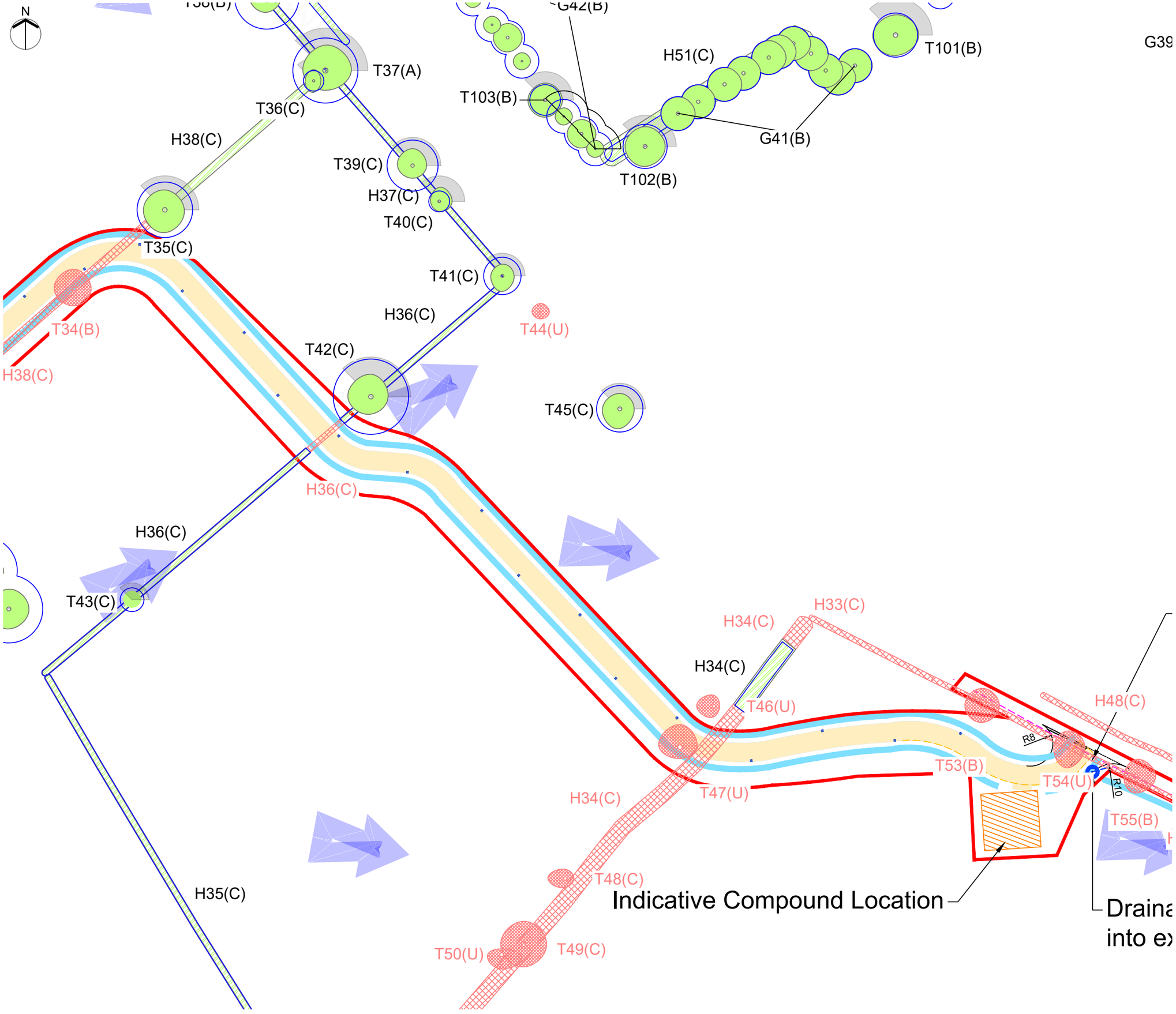


KEY

G39 Tree/Group to be Retained

Tree/Group proposed to be removed subject to relevant permissions

Category U - Unsuitable for retention on arboricultural grounds

Hedgerow Proposed to be Retained and Incorporated into the New Development

Hedgerow proposed to be removed subject to relevant permissions

Root Protection Area (Shown for retained trees only)

T1 (A) G1 (A) Individual / Group Number and BS Category

T1 (A) G1 (A) Individual / Group Number to be Removed and BS 5837:2012 Category

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client
Taylor Wimpey

project
Land North of A420 South Marston and Rowborough, Swindon

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TREE RETENTION PLAN
Haul Road

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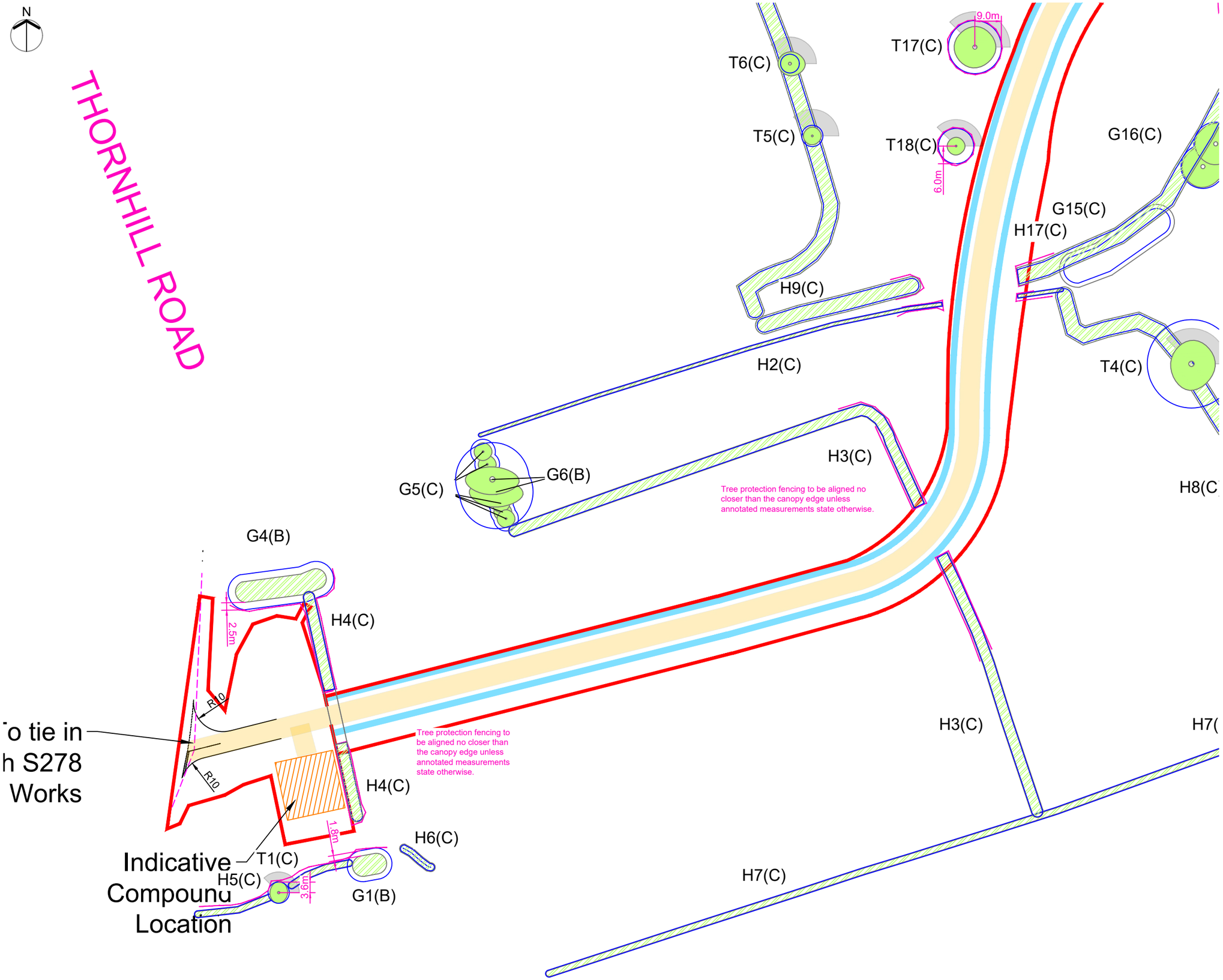


## KEY

- Tree/Group to be Retained
- Hedgerow Proposed to be Retained and Incorporated into the New Development
- Root Protection Area (Shown for retained trees only)
- T1 (A) G1 (A) — Individual / Group Number and BS Category
- Indicative Shade Pattern (in accordance with BS5837:2012 where appropriate)
- 6.0m — Line of Protective Barriers (and distance from tree or retained structure)

Scale 1:1250 @ A3

0 25 50 75m

## NOTES

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Haul Road

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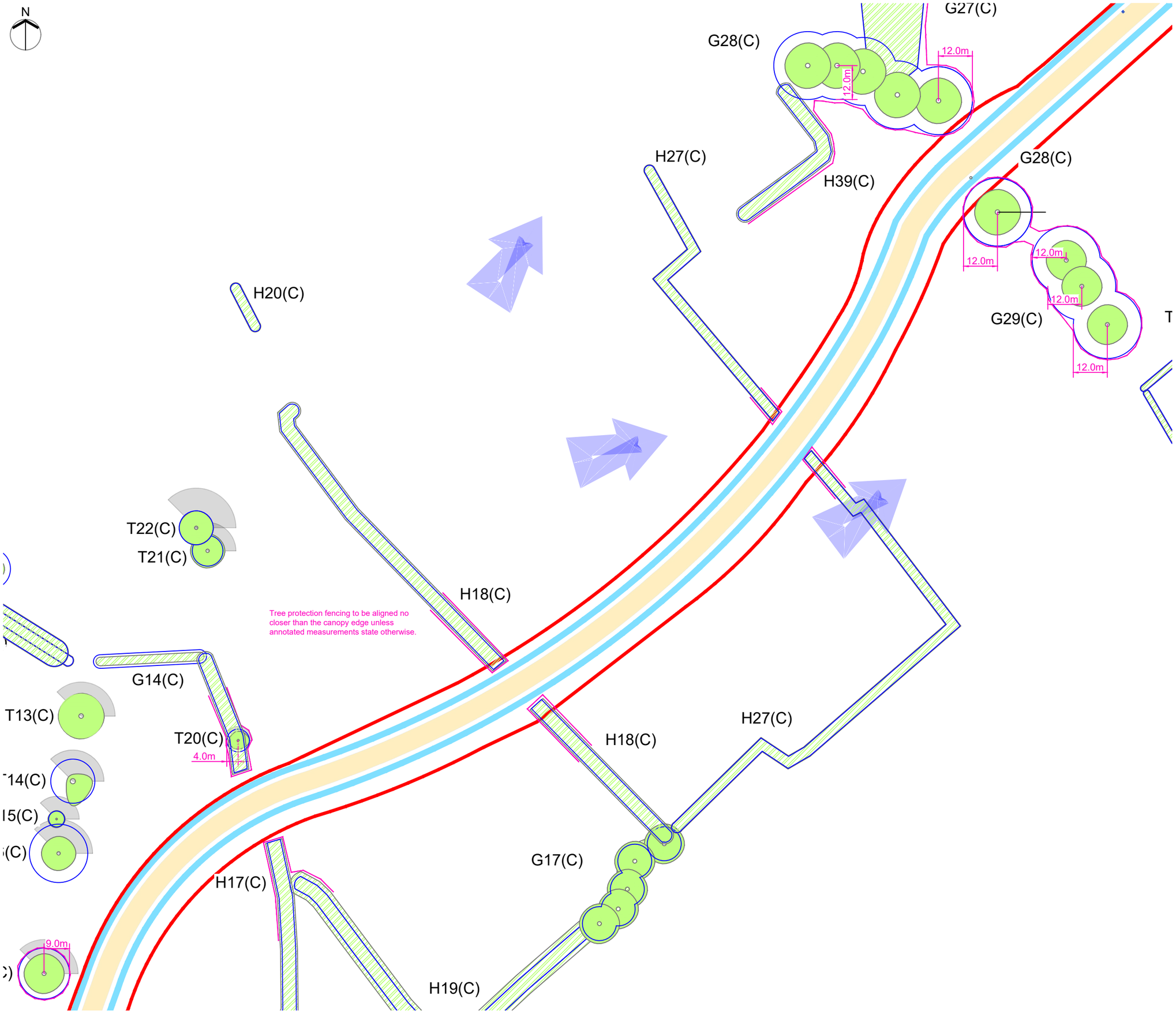


KEY

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Scale 1:1250 @ A3

0 25 50 75m

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Haul Road

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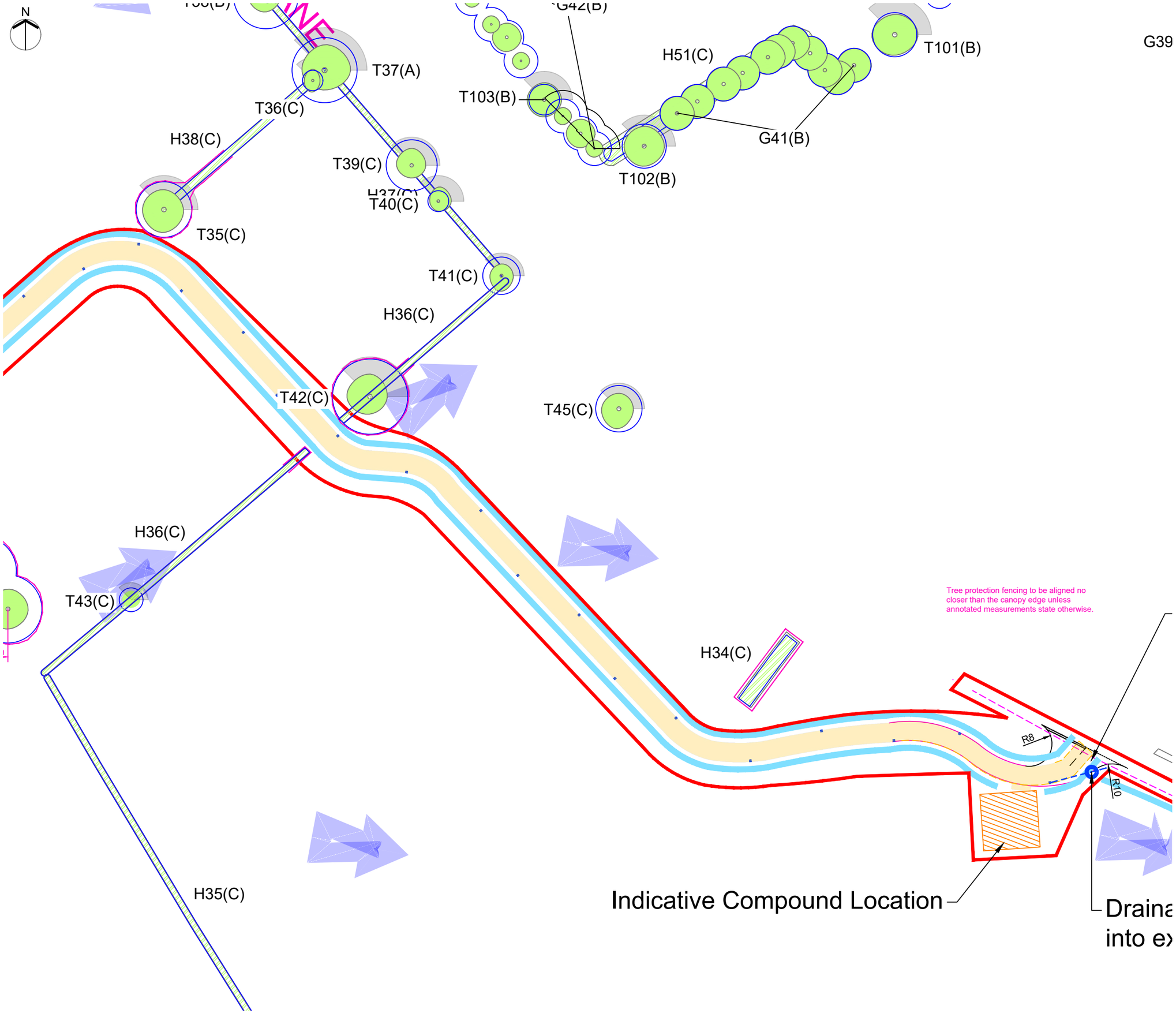


KEY

| | |
|---|---|
| G39 | Tree/Group to be Retained |
| | Hedgerow Proposed to be Retained and Incorporated into the New Development |
| | Root Protection Area (Shown for retained trees only) |
| T1 (A) G1 (A) | Individual / Group Number and BS Category |
| | Indicative Shade Pattern (in accordance with BS5837:2012 where appropriate) |
| 6.0m | Line of Protective Barriers (and distance from tree or retained structure) |

Scale 1:1250 @ A3

0 25 50 75m

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**E**

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# APPENDICES

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# Appendix A - Tree Schedule

| Measurements | Age Classes | Quality Assessment of BS Category | ULE (relates to BS Category) |
|---|---|---|---|
| **Height -** Measured using a digital laser clinometer (m) | **YNG**: Establishing, typically with good vigour and fast growth rates and strong apical dominance; c. less than 1/3 life expectancy | **Category U -** Trees in such a condition that they cannot realistically be retained as living trees in the context of the current land use for longer than 10 years. | <10 years |
| **Stem Dia. -** Diameter measured (mm) in accordance with Annex C of the BS5837 | **SM:** Semi-mature trees less than 1/3 life expectancy | **Category A -** Trees of high quality with an estimated remaining life expectancy of at least 40 years. | 40+ years |
| **Crown Radius -** Measured using a digital laser clinometer radially from the main stem (m) | **EM:** Established, typically vigorous and increasing in apical height and lateral spread; 1/3 - 2/3 life expectancy. Offers landscape significance | **Category B -** Trees of moderate quality with an estimated remaining life expectancy of at least 20 years. | 20-40 years |
| **Abbreviations**<br><br>est - Estimated stem diameter<br>avg - Average stem diameter for multiple stems<br>upto - Maximum stem diameter of a group | **M:** Fully established over 2/3 life expectancy, generally good vigour and achieving full height potential with crown still spreading | **Category C -** Trees of low quality with an estimated remaining life expectancy of at least 10 years, or young trees with a stem diameter below 150mm. | 10-20 years |
| | **OM:** Fully mature, at the extremes of expected life expectancy, vigour decreasing, declining or moribund | Sub-categories: (i) - Mainly arboricultural value<br>(ii) - Mainly landscape value<br>(iii) - Mainly cultural or conservation value | |
| | **V:** biological, cultural or aesthetic value comprising niche saproxylic habitat. Individuals of large proportions (stem girth) in comparison to trees of the same species/surviving beyond the typical age range for their species. | **The BS category particular consideration has been given to the following:**<br>• The presence of any structural defects in each tree/group and its future life expectancy<br>• The size and form of each tree/group and its suitability within the context of a proposed development<br>• The location of each tree relative to existing site features e.g. its screening value or landscape features<br>• Age class and life expectancy | |

| Structural Condition | Physiological Condition |
|---|---|
| **Good -** No significant structural defects | **Good -** No significant health problems |
| **Fair -** Structural defects that can be remediated | **Fair -** Symptoms of ill-health that can be remediated |
| **Poor -** Significant defects beyond remediation, present a risk of failure in the foreseeable future | **Poor -** Significant ill-health. Unlikely the tree will recover in the long term |
| **Dead -** Dead tree with structural integrity of tree severely compromised | **Advanced Decline / Dead -** Advanced state of decline and unlikely to recover or Dead |

| Root Protection Area (RPA) |
|---|
| • The RPA Radius column provides the extent of an equivalent circle from the centre of the stem (m).<br><br>• The RPA is calculated using the formulae described in paragraph 4.6.1 of British Standard 5837: 2012 and is indicative of the rooting area required for a tree to be successfully retained. Tree roots extend beyond the calculated RPA in many cases and where possible a greater distance should be protected.<br><br>• Where veteran trees have been identified the RPA has been calculated in accordance with Natural England guidance i.e. 15x the stem diameter, uncapped. |

## Appendix Summary

| | Individual Trees | Totals | Tree Groups and Hedgerows | Totals |
|---|---|---|---|---|
| **Category U** | T3, T33, T44, T46, T47, T50, T54, T66, T68, T104, T106, T107 | 12 | G3 | 1 |
| **Category A** | T37, T65, T78 | 3 | | 0 |
| **Category B** | T21, T34, T38, T53, T55, T56, T57, T59, T62, T67, T73, T74, T75, T76, T77, T102, T205 | 17 | G1, G4, G6, G13, G32, G41, G42 | 7 |
| **Category C** | T4, T5, T6, T13, T14, T15, T16, T17, T18, T19, T20, T22, T35, T36, T39, T40, T41, T42, T43, T45, T48, T49, T51, T58, T60, T61, T63, T64, T79, T105, T204 | 31 | G5, G11, G12, G14, G15, G16, G17, G26, G27, G28, G29, G30, G31, G34, G35, G66, H1, H2, H3, H4, H5, H6, H7, H8, H9, H10, H12, H16, H17, H18, H19, H20, H27, H28, H29, H30, H31, H32, H33, H34, H35, H36, H37, H38, H39, H40, H41, H47, H48, H49, H51, H88 | 52 |
| | **Total** | 63 | **Total** | 60 |